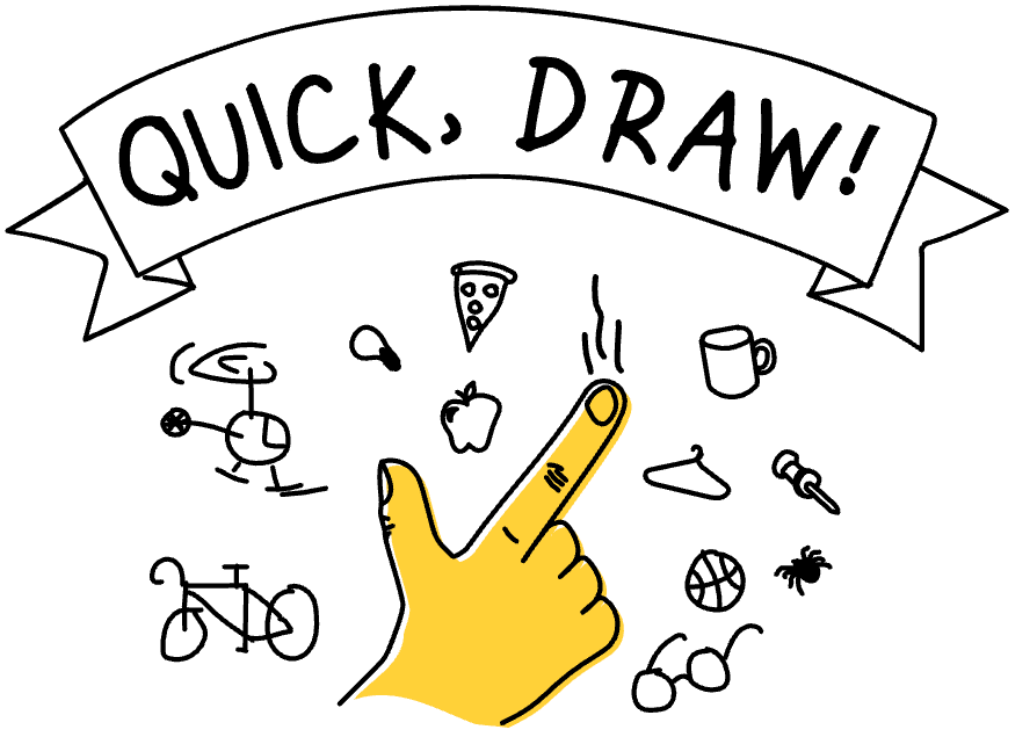


上海科技大学
ShanghaiTech University

Edge Detection

CS172 Computer Vision I

Instructor: Jiayuan Gu



Can a neural network learn to recognize doodling?

Help teach it by adding your drawings to the world's largest doodling data set, shared publicly to help with machine learning research.

https://quickdraw.withgoogle.com/

You were asked to draw bee

You drew this, and the neural net didn't recognize it.



It thought your drawing looked more like these:

Closest match
sea turtle



2nd closest match
mouse



3rd closest match
shark



https://blog.tcea.org/quick-draw-google/

Agenda

- Roadmap
  - Image gradients
  - Derivatives with convolution
  - Gaussian derivative filter
  - Sobel operator
  - Canny edge detection

- Objectives
  - compute/interpret image gradients
  - explain smoothing–localization trade-off
  - explain the Canny pipeline

Edge Detection



Winter in Kraków photographed by Marcin Ryczek

Edge Detection

- **Goal:** Identify sudden changes (**discontinuities**) in an image

- Why do we study edge detection?
  - Intuitively, semantic and shape information from the image can be encoded in the edges
  - More compact than pixels

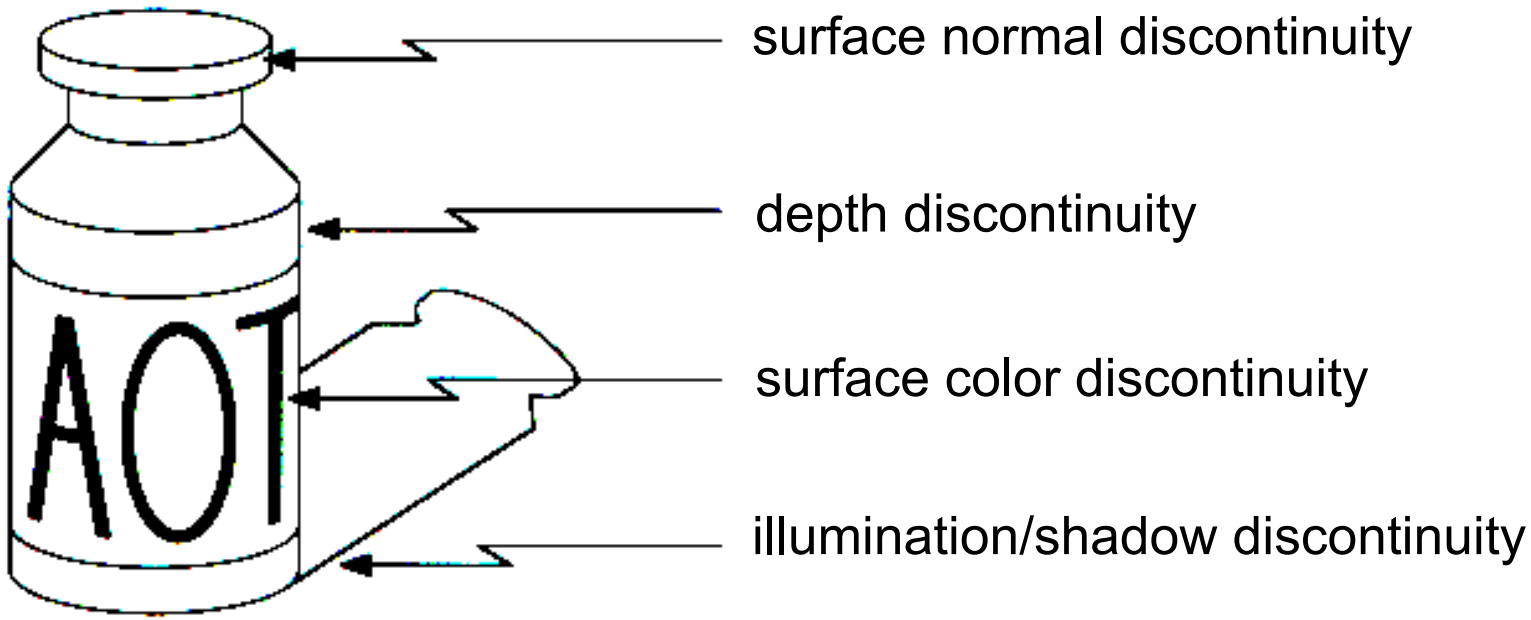


artist's line drawing is an example of edges

Source: D. Lowe

Origin of Edges

Edges are caused by a variety of factors:



Source: Steve Seitz

Characterizing Edges

- An edge is a place of rapid change in the image intensity function

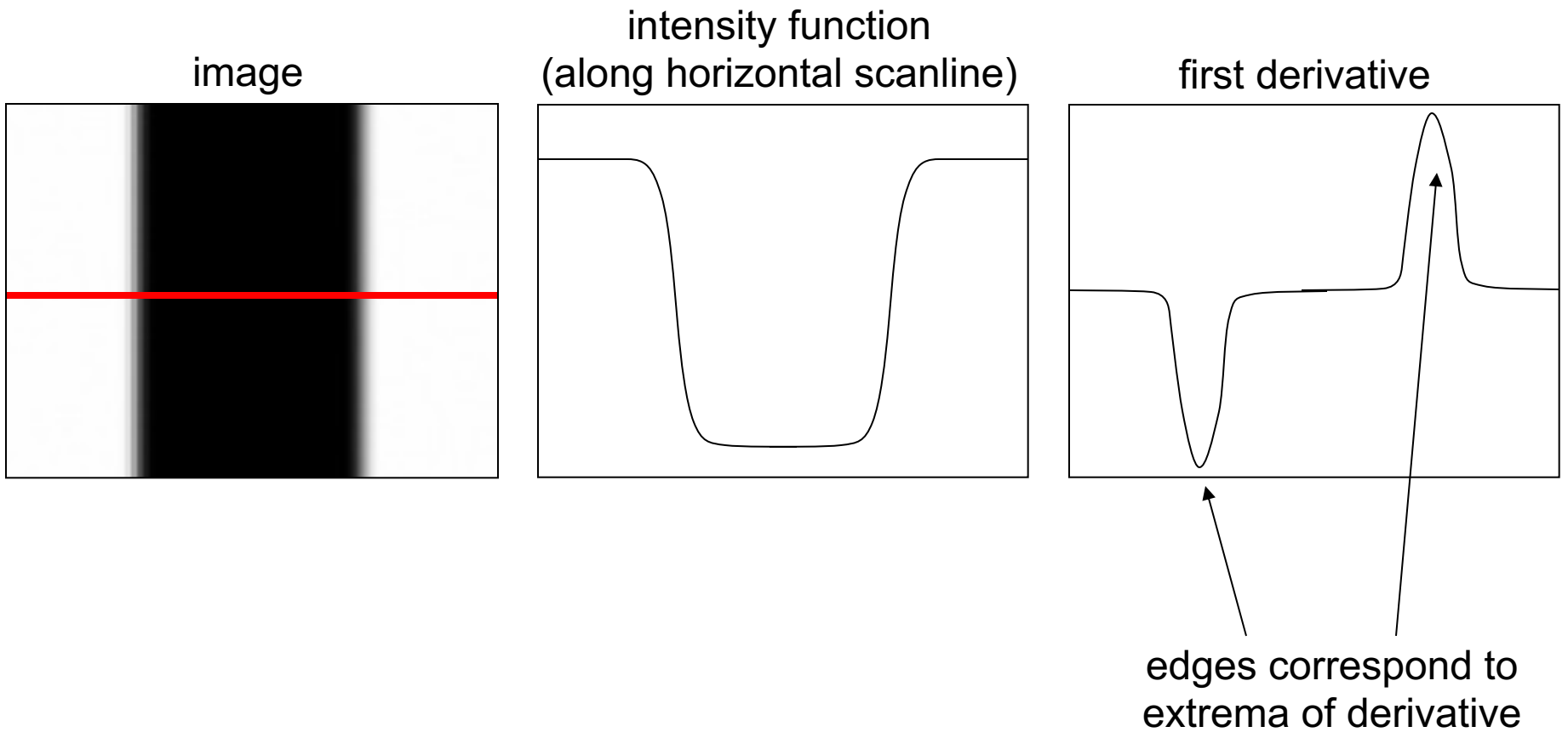


Image Derivatives

For a 2D image $f(x, y)$, the partial derivative is:

$$\frac{\partial f(x,y)}{\partial x} = \lim_{\varepsilon \to 0} \frac{f(x+\varepsilon, y) - f(x,y)}{\varepsilon}$$

For discrete data, we can approximate using finite differences:

$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1, y) - f(x,y)}{1}$$

To implement the above as **convolution**, what would be the associated **filter**?

Source: K. Grauman

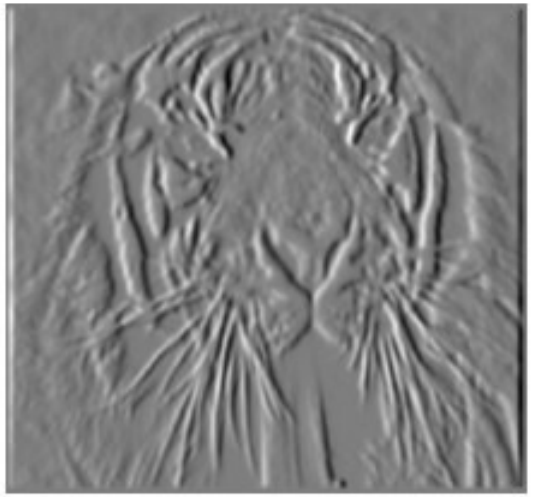
Partial Derivatives of an Image

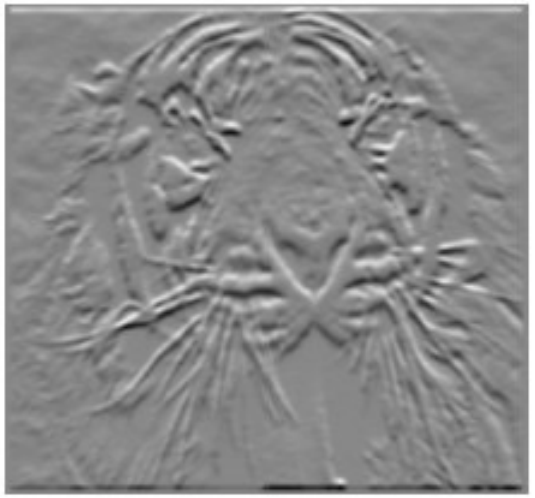
$$\frac{\partial f(x,y)}{\partial x}$$
(usually denoted by I_x)



$$\frac{\partial f(x,y)}{\partial y}$$
(usually denoted by I_y)

-1	1





-1
1

Which shows changes with respect to x?

The kernel is already flipped for simplicity

Image Gradient

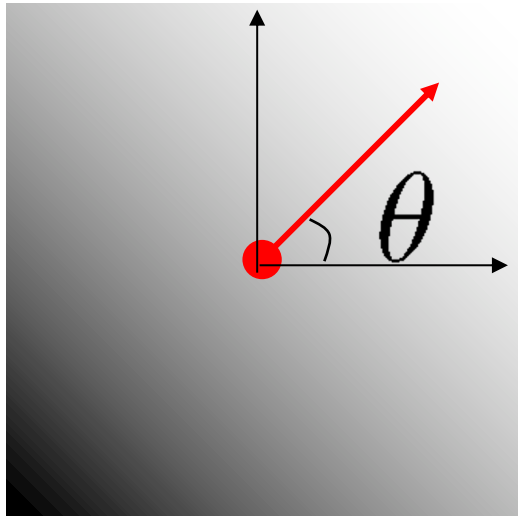


$$\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$



Gradient magnitude: $\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$

Image Gradient



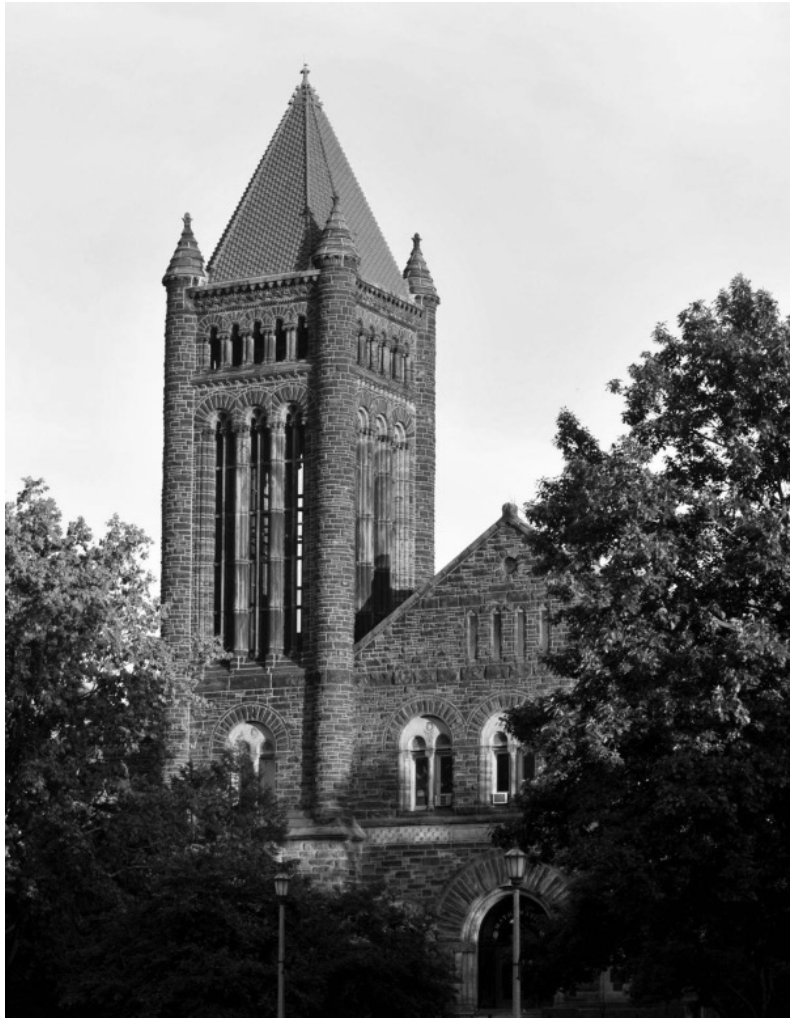
$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

Gradient direction: $\theta = \tan^{-1}(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x})$

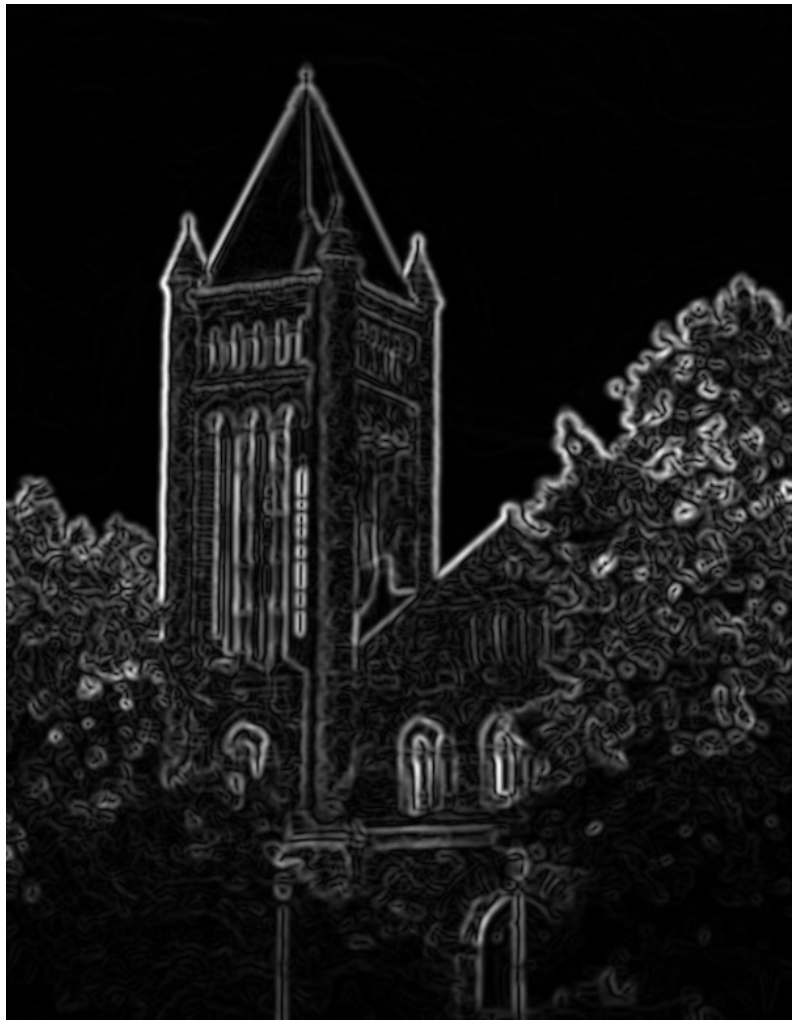
Gradient direction is perpendicular to the edge

Source: Steve Seitz

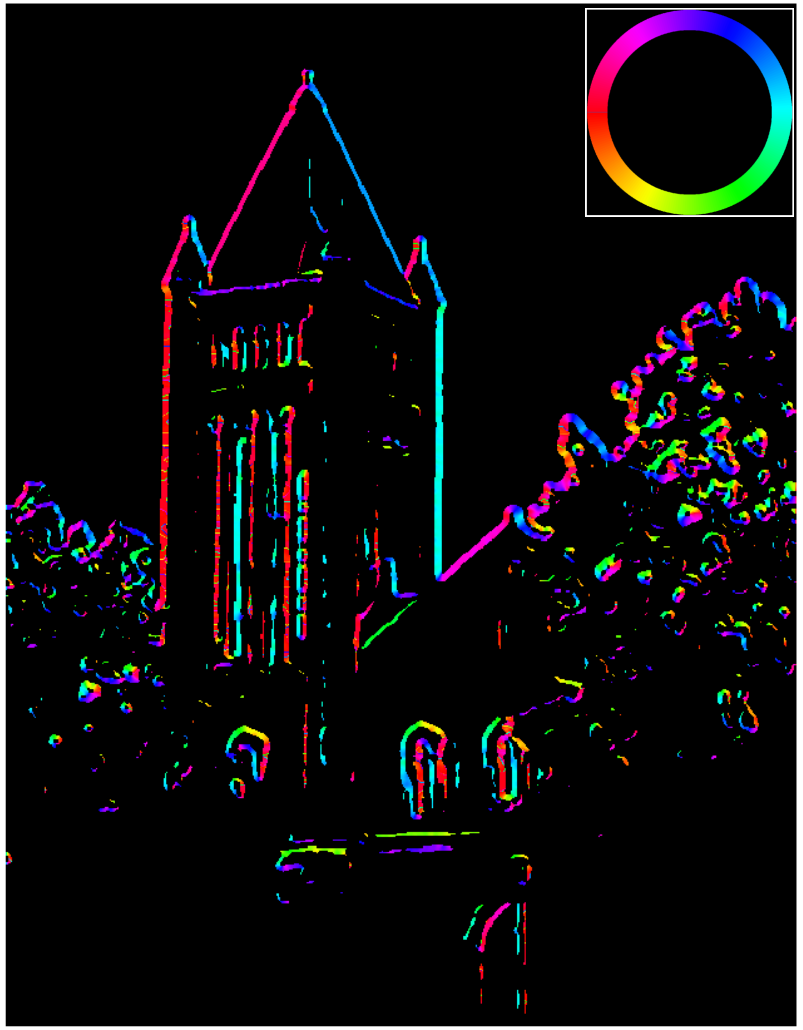
Image Gradient



original image



gradient norm



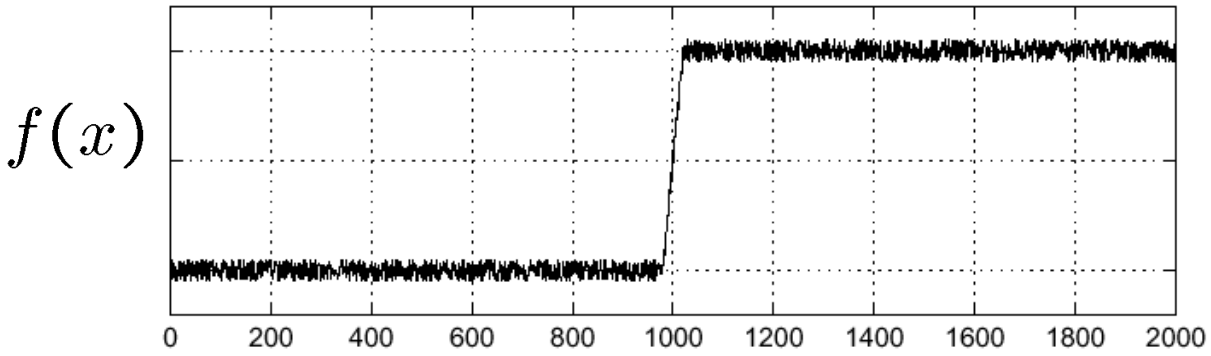
gradient orientation

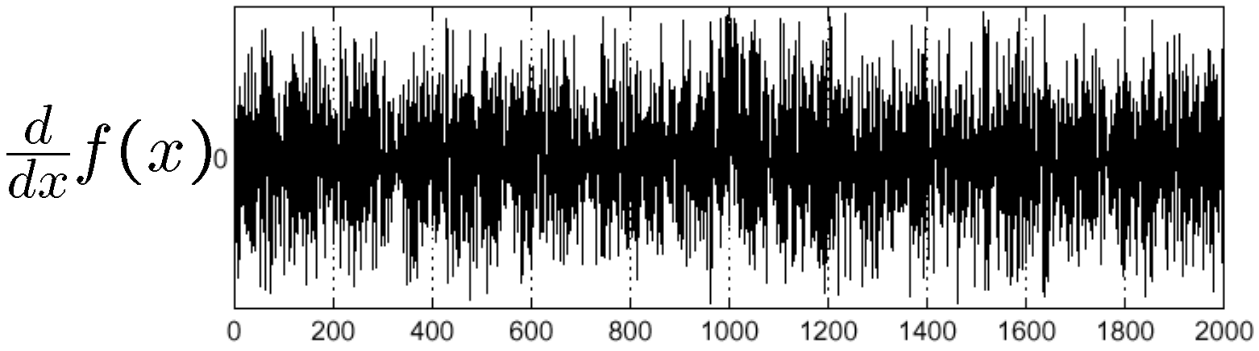
Source: Saurabh Gupta

Gaussian Derivative Filter

Effects of Noise

- Consider a single row or column of the image



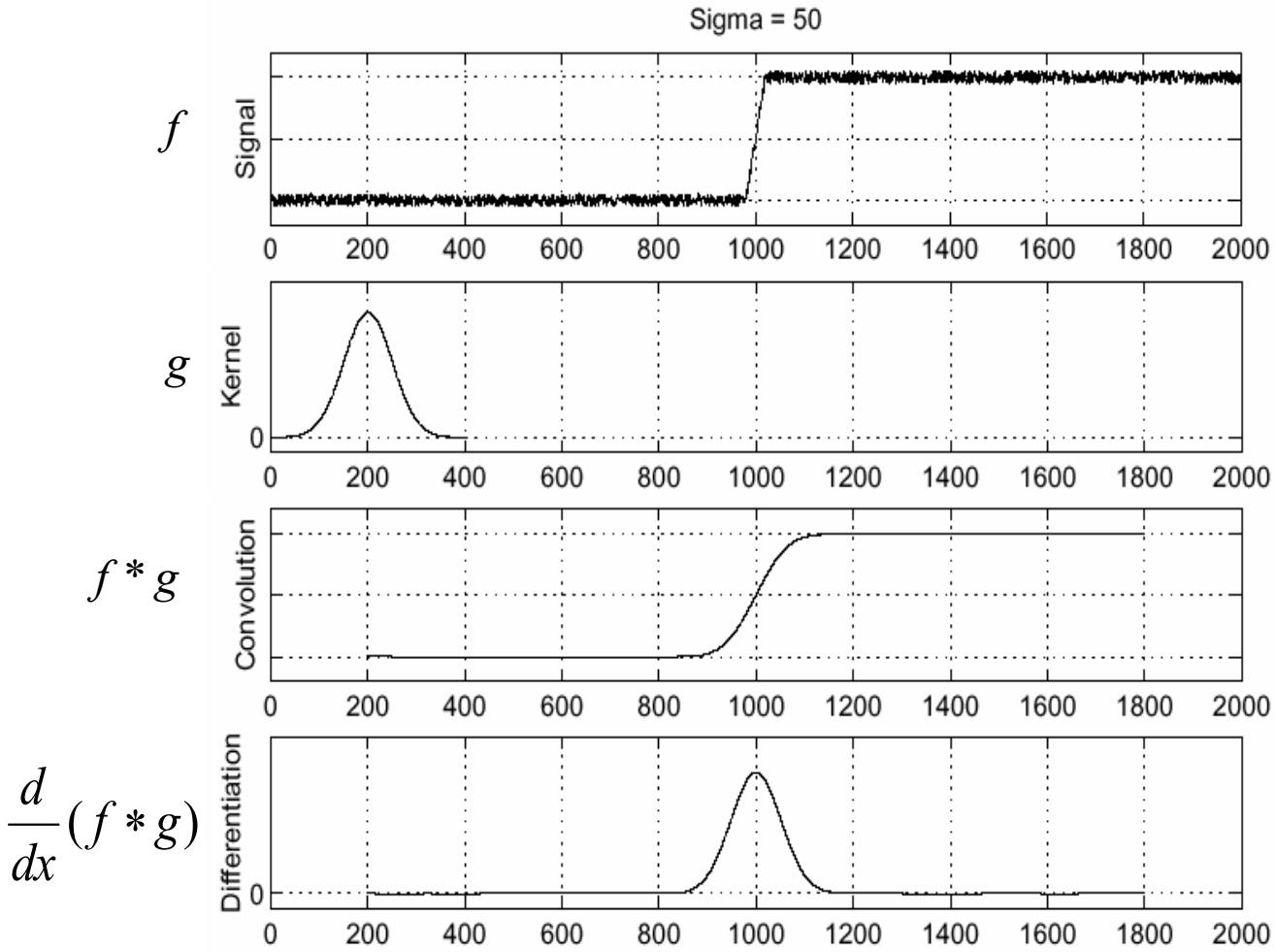


Where is the edge?

Differentiation amplifies rapid fluctuations, including noise

Source: S. Seitz

Solution: Smooth First



To find edges, look for peaks in $\frac{d}{dx}(f * g)$

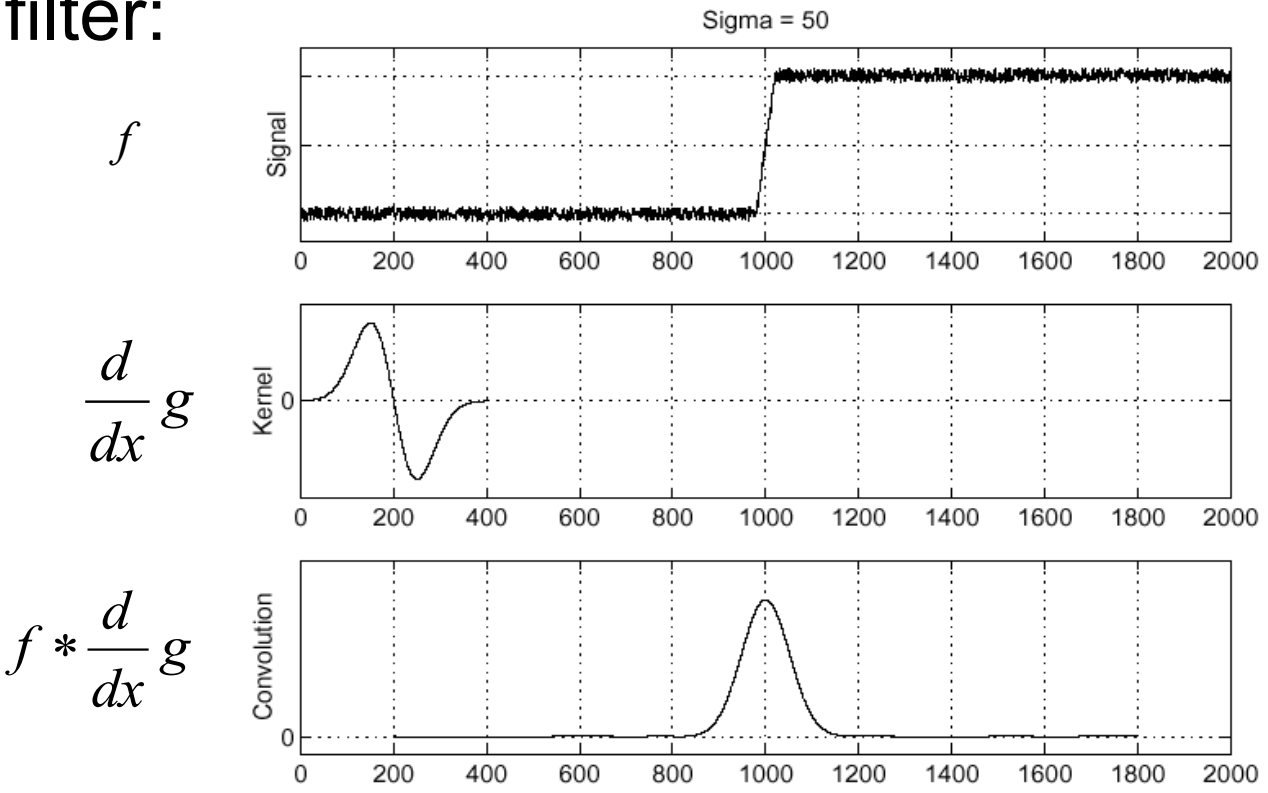
Source: S. Seitz

Derivative Theorem of Convolution

- Differentiation is convolution, and convolution is associative:

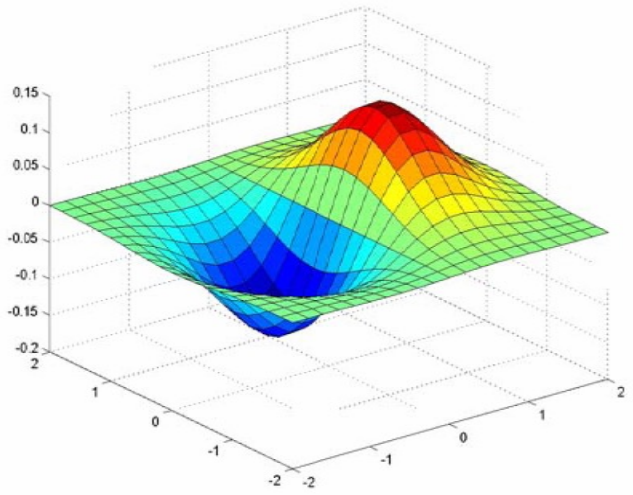
$$\frac{d}{dx}(f * g) = f * \frac{d}{dx} g$$

- Smoothing and differentiation can be combined into one derivative-of-Gaussian filter:

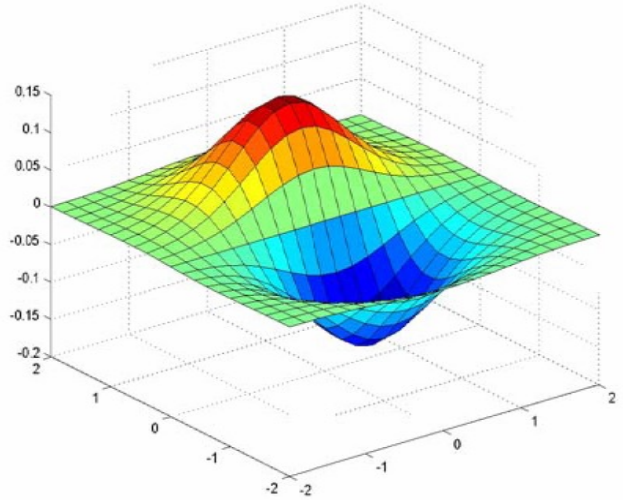


Source: S. Seitz

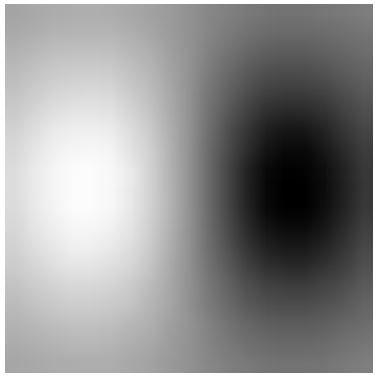
Derivative of Gaussian Filters

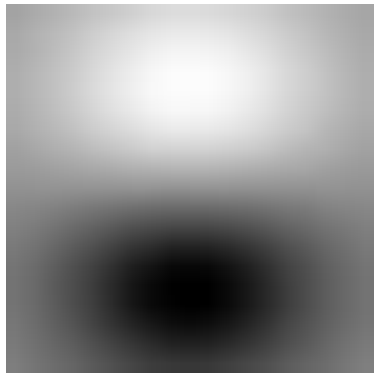


x-direction



y-direction

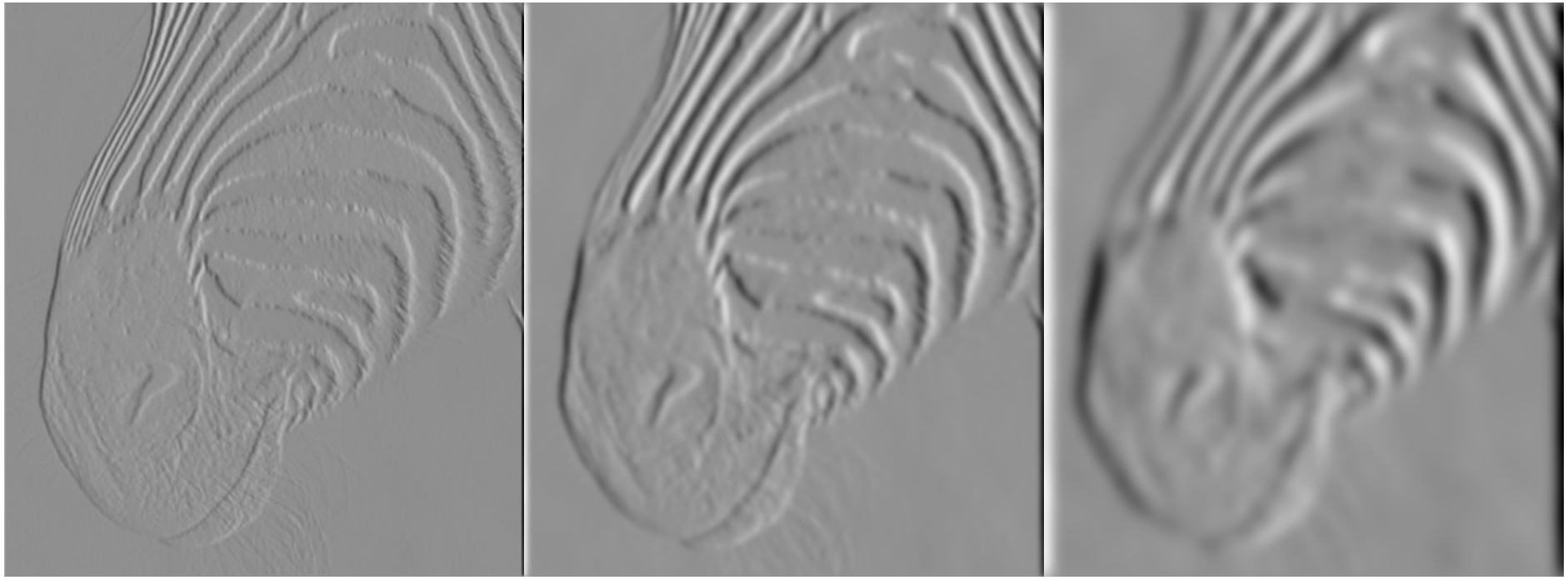




Which one finds horizontal/vertical edges?

Are these filters separable?

Scale of Gaussian Derivative Filter



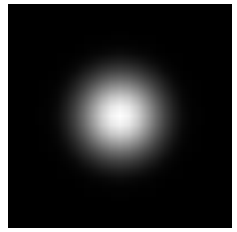
$\sigma = 1$ pixel | 3 pixels | 7 pixels

Smoothed derivative removes noise, but blurs edge.
Also finds edges at different “scales”

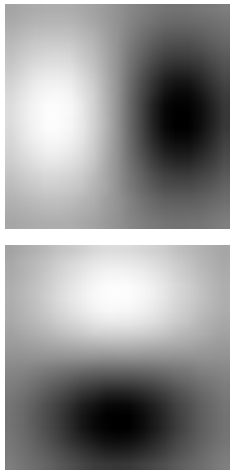
Source: D. Forsyth

Smoothing vs. Derivative Filters

- Smoothing filters
  - Gaussian: attenuate “high-frequency” components; “low-pass” filter
  - Can the values of a smoothing filter be negative?
  - What should the values sum to?
    - **One:** constant regions are not affected by the filter



- Derivative filters
  - Derivatives of Gaussian
  - Can the values of a derivative filter be negative?
  - What should the values sum to?
    - **Zero:** no response in constant regions



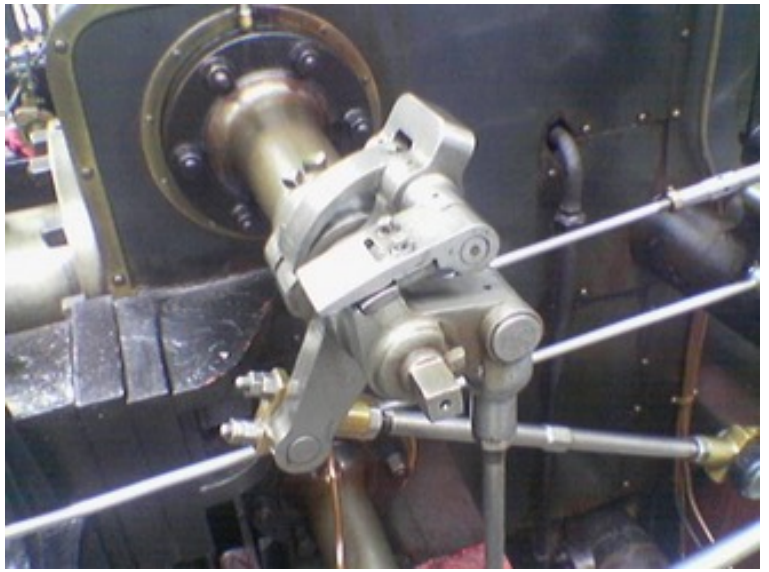
Gradient-Based Edge Detection

Sobel operator and Canny edge detection

Sobel Operator

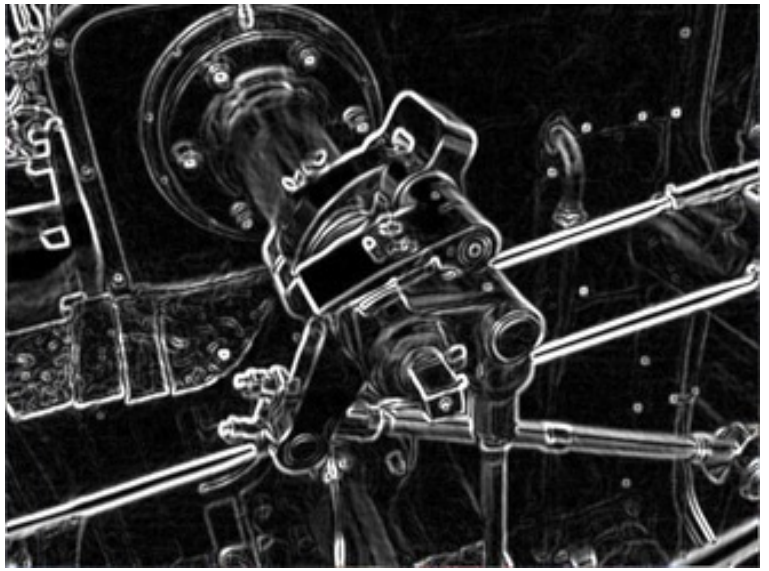
$$S_x = [1,2,1]^T[-1,0,1] = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

smooth vertically differentiate horizontally Ignoring normalization



An example image from wiki

$$S_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$



The Sobel operator applied to that image

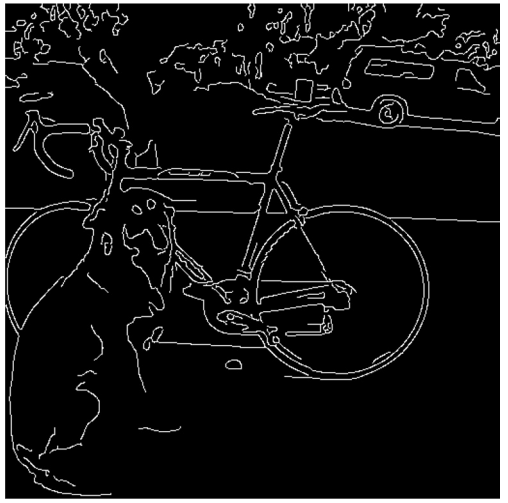
The kernel is already flipped for simplicity

Canny Edge Detection

Algorithm:

- 1. Smooth image (only want “real” edges, not noise)
- 2. Calculate gradient direction and magnitude
- 3. Non-maximum suppression perpendicular to edge
- 4. Threshold into strong, weak, no edge
- 5. Track edges by hysteresis





J. Canny, A Computational Approach To Edge Detection, IEEE Trans. Pattern Analysis and Machine Intelligence, 8:679-714, 1986.

Credit to: UW CSE455 Linda Shapiro

Canny Edge Detection: Smooth

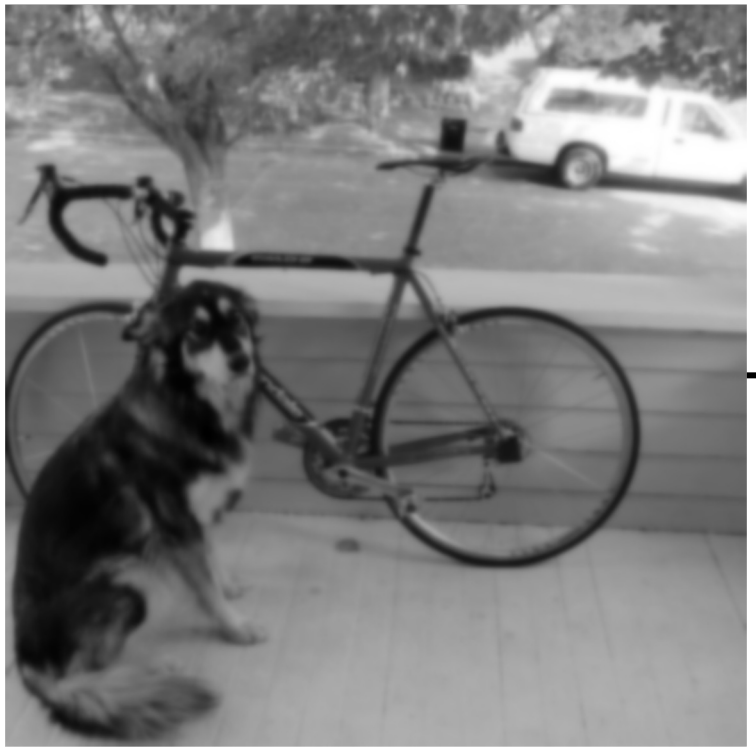


Gaussian filter



Credit to: UW CSE455 Linda Shapiro

Canny Edge Detection: Gradient



Sobel operator

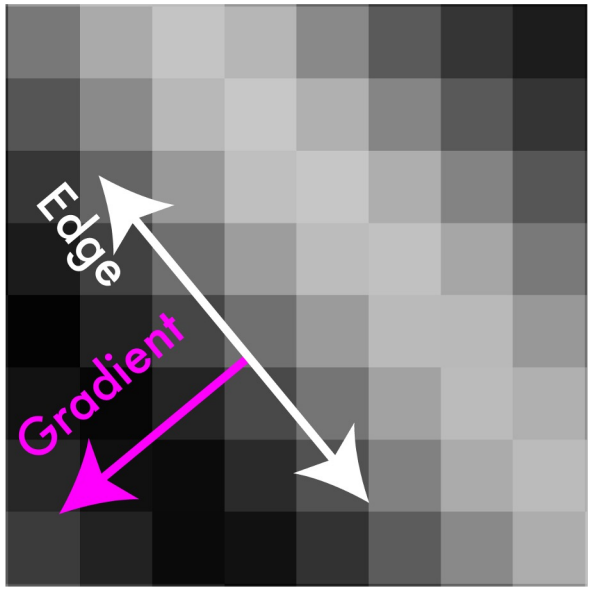


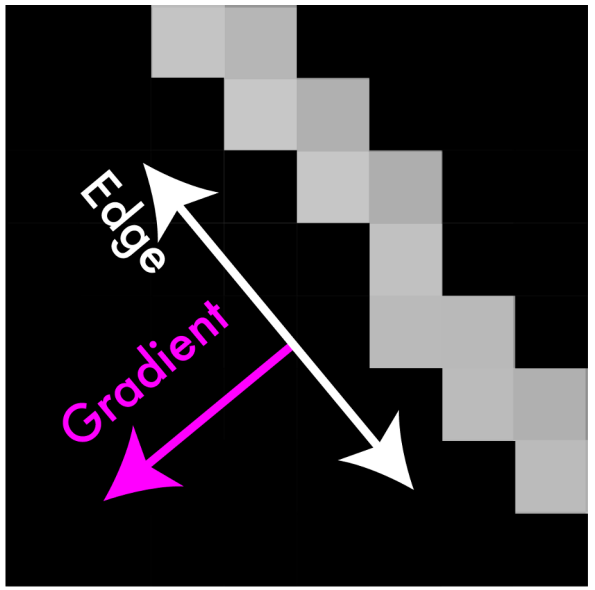
Credit to: UW CSE455 Linda Shapiro

Canny Edge Detection: Non-Maximum Suppression



- Want single pixel edges, not thick blurry lines
- Need to check nearby pixels
- See if response is highest





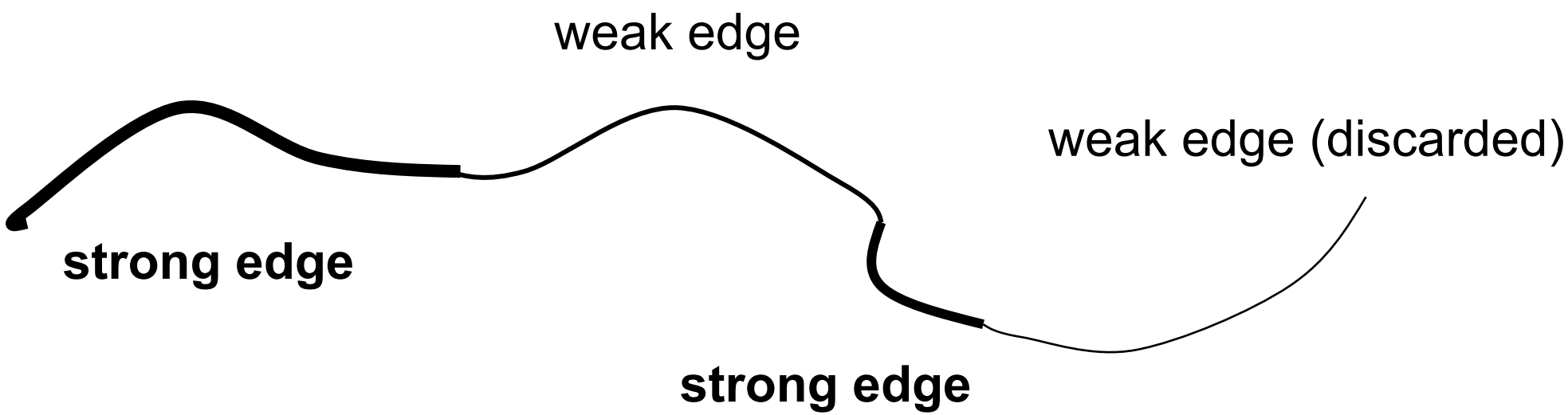
Credit to: UW CSE455 Linda Shapiro

Canny Edge Detection: Double Thresholding

- There is still noise, because we have marked maxima of the gradient magnitude without regard to how large these maxima are
  - Solution: Apply a threshold test to ensure that the maxima are greater than some lower bound
  - However, it might lead to broken edge curves

- The usual trick for dealing with this is to use **hysteresis**
  - introducing **two thresholds**: T_H and T_L
  - Categorize edges (or points/pixels) into
    - strong: $> T_H$
    - weak: $T_L < x < T_H$
    - no: $< T_L$

Track Edges by Hysteresis Thresholding



Use a high threshold to start edge curves, and a low threshold to continue them

Source: Steve Seitz

Hysteresis Thresholding



original image



high threshold
(strong edges)



low threshold
(weak edges)



hysteresis threshold

Source: L. Fei-Fei

Summary

- Edge is rapid image change
- Image gradient includes both magnitude and direction
- Smoothing scale controls noise/detail
- Canny edge detection: smooth → gradient → NMS → thresholds → hysteresis

Reading List

- **Szeliski, *Computer Vision: Algorithms and Applications*, 2e**
  - 3.2 *Linear Filtering*; 7.2.1 *Edge Detection*
- **Forsyth & Ponce, *Computer Vision: A Modern Approach*, 2e**
  - 5.1–5.2 *Image Gradients & Edge Detectors*; 6.1 *Texture Using Filters*
- J. Canny, **"A Computational Approach to Edge Detection,"** TPAMI 1986
- OpenCV, **Canny Edge Detector Tutorial**
- https://bigwww.epfl.ch/demo/ip/demos/edgeDetector/
- P. Perez, M. Gangnet, A. Blake, Poisson Image Editing, SIGGRAPH 2003